

University Academic Analytics Power BI Project Report

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Course Title: Data Visualization

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Data Analysis and Reporting Using Power BI University Academic Analytics

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Executive Summary

This project presents a complete university academic analytics solution using synthetic data and Power BI-ready modeling assets. The work includes data generation, raw and cleaned datasets, validation, relationship documentation, DAX and Power Query libraries, a public analytics viewer, a demonstration video, and a GitHub release. The cleaned dataset contains 1,000 students, 10 departments, 70 courses, 50 faculty members, 12,500 enrollment records, 22,577 attendance records, and 5,994 payment records. Power BI Desktop was unavailable, so the report honestly provides build-ready assets and a static analytics publication rather than a fabricated PBIX.

Introduction

Data visualization helps academic administrators interpret complex institutional activities such as enrollment, grades, attendance, tuition collection, scholarship distribution, and budget utilization. This project applies a structured data-analysis workflow to a fictional university context and demonstrates how related operational tables can be transformed into a management-ready analytics system.

Problem Statement

University decision makers often need to combine information from student records, course registration, attendance, finance, and departmental budgeting. Without a consistent data model, reporting becomes fragmented and difficult to validate. This project addresses that problem through a controlled relational dataset, automated validation, documented transformations, and a published analytics interface.

Objectives

The objectives are to generate realistic synthetic data; preserve separate raw and cleaned datasets; demonstrate one-to-one, one-to-many, and bridge-resolved many-to-many relationships; document a Power BI semantic model; define DAX measures and calculated columns; specify dashboard pages; validate data quality; publish a reviewer-friendly analytics viewer; and provide formal academic documentation.

Methodology

The project followed a reproducible pipeline: synthetic data generation, controlled raw-data issue injection, cleaning, validation, documentation, analytics publication, testing, and GitHub release management. Python was used for data generation, cleaning, validation, report-support files, and website data preparation. GitHub Pages was used to publish the static analytics viewer.

Dataset Description

The dataset covers academic years 2023 to 2026 and uses fictional student, faculty, course, attendance, payment, and budget records. It contains ten required tables: Students, StudentProfile, Departments, Courses, Faculty, Semesters, Enrollment, Attendance, Payments, and DepartmentBudget. No real student or private institutional information is included.

Table 1. Dataset table summary

Data Cleaning and Transformation

The raw dataset intentionally contains duplicates, inconsistent capitalization, leading/trailing spaces, invalid marks, null attendance values, text dates, and inconsistent payment labels. Cleaning standardizes text, removes duplicates, converts dates and numeric fields, clamps marks to valid ranges, recalculates payment balances, fills scholarship status correctly, and creates analytical risk categories.

Data Model and Relationships

The design uses dimension-style tables for Students, StudentProfile, Departments, Courses, Faculty, and Semesters. Enrollment acts as both a fact table and bridge table between Students and Courses. Attendance, Payments, and DepartmentBudget are fact tables. Single-direction filtering is recommended where possible to reduce ambiguity.

DAX Measures and Calculations

The project includes 40 documented DAX measures covering counts, pass/fail rates, attendance risk, payment collection, outstanding amounts, faculty load, growth rates, budget utilization, scholarship rate, ranking, dynamic titles, TOPN, CONCATENATEX, and USERRELATIONSHIP examples. Date table DAX and calculated-column examples are also provided.

Dashboard Development

The published interface follows the visual direction of a dark executive dashboard with gold highlights, compact KPI cards, high-contrast panels, and responsive chart containers. The website includes KPI cards, line charts, horizontal bar charts, a doughnut chart, budget table, analytical findings, download links, screenshots, and a demo

video.

Implementation Process

Implementation was organized into scripts, data folders, documentation folders, report files, presentation files, screenshots, website source, and release assets. The published site in docs/ reads generated JSON analytics data from the cleaned workbook, ensuring that the displayed figures match the validated dataset.

Testing and Validation

Automated validation checks tested primary-key uniqueness, foreign-key integrity, marks, grade points, attendance percentages, payment calculations, date logic, duplicate identifiers, orphan records, and required columns. The website was tested through direct HTTP requests, desktop screenshots, mobile screenshots, and download-link verification.

Table 2. Validation summary

Findings and Interpretation

The analysis shows enrollment concentration by department, high-volume semester patterns, variation in average marks by department, attendance-performance association, outstanding tuition exposure, budget utilization differences, and scholarship distribution. Findings are interpreted as operational indicators, not causal claims.

Table 3. Main analytical findings

Limitations

The dataset is synthetic and cannot be used to infer real university behavior. Power BI Desktop was not installed, so the PBIX file and Power BI Service report were not created. The public analytics viewer is static and does not replace a live Power BI Service report. Further refinement should occur after building the actual PBIX.

Recommendations

Future work should build and publish the PBIX in Power BI Desktop, add real Power BI screenshots, configure data refresh, enhance drill-through pages, include accessibility testing with screen readers, and refine measures after stakeholder feedback. Academic teams should use attendance and payment risk categories as early-warning indicators.

Conclusion

The final submission provides a comprehensive, validated, and publicly accessible analytics project package. It demonstrates relational modeling, data cleaning, validation, DAX planning, Power Query documentation, reporting design, public deployment, and academic documentation while clearly distinguishing completed artifacts from environment-limited Power BI tasks.

References

Microsoft. (2026). Power BI documentation. Microsoft Learn.

Microsoft. (2026). DAX function reference. Microsoft Learn.

Microsoft. (2026). Power Query M language specification. Microsoft Learn.

GitHub. (2026). GitHub Pages documentation. GitHub Docs.

Python Software Foundation. (2026). Python documentation.

Project dataset and visualizations are synthetically generated by the author for academic demonstration.

Appendices

Appendix A: DAX_Measures.md. Appendix B: Power_Query_M_Code.md. Appendix C: Date_Table_DAX.md. Appendix D: Relationship_Documentation.md. Appendix E: Prompt_Compliance_Cross_Check.md. Repository: <https://github.com/Monokayser/University-PowerBI-Analytics>. Published viewer: <https://monokayser.github.io/University-PowerBI-Analytics/>. Demo: <https://monokayser.github.io/University-PowerBI-Analytics/demo.html>. Release: <https://github.com/Monokayser/University-PowerBI-Analytics/releases/tag/v1.1.0>.

Figure 1. Enrollment trend by semester

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Figure 2. Students by department

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Figure 3. Average final mark by department

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Figure 4. Outstanding fees by department

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Power BI Publication Limitation and Academic Integrity Note

Power BI Desktop and Power BI Service publishing access were not available in the automated build environment used for this repository update. For this reason, the project does not claim a fabricated .pbix file, Power BI Service URL, app URL, public embed link, or authenticated service screenshot. The repository instead provides the cleaned workbook, DAX measure library, Power Query guidance, relationship documentation, dashboard specifications, a static published analytics viewer, and verification screenshots. These materials allow the report to be reproduced accurately in Power BI Desktop when the required Microsoft Power BI environment is available.

Verification evidence: the static analytics viewer, downloadable report, cleaned dataset, screenshots, demo video, and release package were tested through the local site and GitHub Pages workflow. Dataset validation passed with no failed validation rules.

Table 1. Report table extract

Table	Rows	Columns
Students	1,000	12
StudentProfile	1,000	7
Departments	10	6
Courses	70	7
Faculty	50	7
Semesters	11	5
Enrollment	12,500	14
Attendance	22,577	6
Payments	5,994	10
DepartmentBudget	40	11

Table 2. Report table extract

Result	Checks	Failed Checks	Evidence
Passed	36	0	data/validation/Data_Validation_Results.xl

Table 3. Report table extract

Finding	Observation	Recommended Action	Limitation
Highest enrollment department	English has the largest student count.	Review section sizes and faculty allocatio	Synthetic dataset only.
Highest failure course	C018 has the highest failure rate.	Add tutoring and review course design.	Association only, not causation.
Attendance relationship	Lower attendance bands show lower average	Trigger advising below 75% attendance.	Cannot prove attendance causes marks.
Outstanding payments	Outstanding balance totals 125,733,250.	Segment outreach by payment risk level.	Synthetic tuition assumptions.
Budget utilization	D007 has the highest utilization in 2023.	Review mid-year budget reallocations.	Budget is generated, not actual finance da
Scholarship distribution	32.0% of students receive a scholarship.	Track outcomes by scholarship type.	No real socioeconomic data.
Faculty workload	Enrollment links faculty to taught courses	Monitor distinct courses and enrollments p	Workload excludes non-teaching duties.
Graduation eligibility	Credit and GPA measures identify students	Add drill-through student review page.	Eligibility rule is simplified.
Payment status	Paid, partial, and unpaid categories provi	Use respectful outreach workflows.	Synthetic data includes fictional profiles
Semester trend	Semester fields support enrollment and per	Monitor semester-over-semester growth.	Trend is generated from random distributio

Appendix A. Analytical Figures and Diagrams

Figure 1: Enrollment Trend by Semester

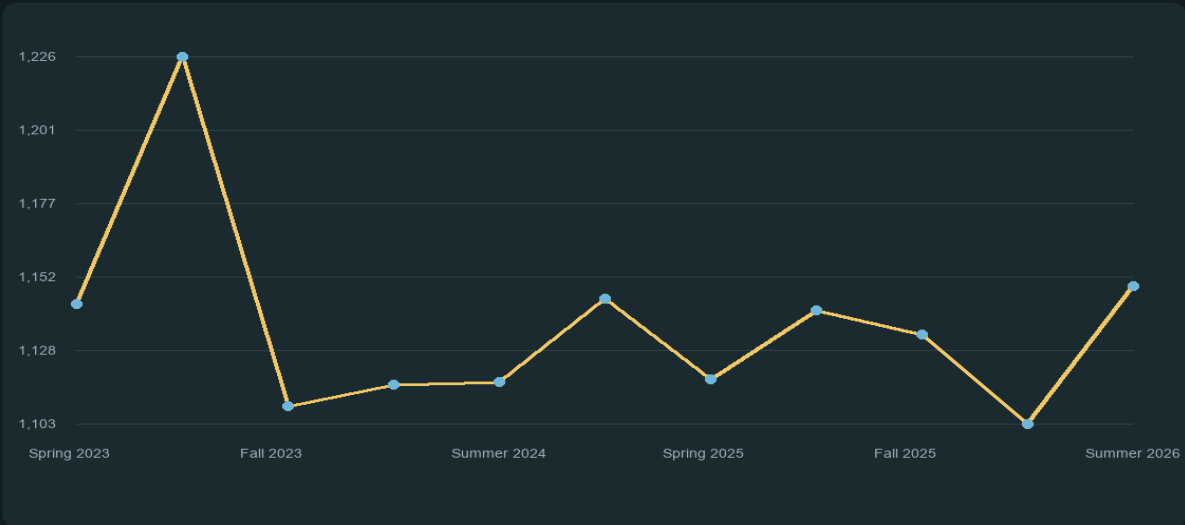


Figure A1. Enrollment trend by semester.

Figure 2: Students by Department



Figure A2. Student distribution by department.

Figure 3: Average Final Mark by Department



Figure A3. Average final mark by department.

Figure 4: Outstanding Fees by Department



Figure A4. Outstanding balance by department.

Figure 5: Budget Utilization by Department



Figure A5. Budget utilization comparison.

Figure 6: System Architecture and Publication Workflow

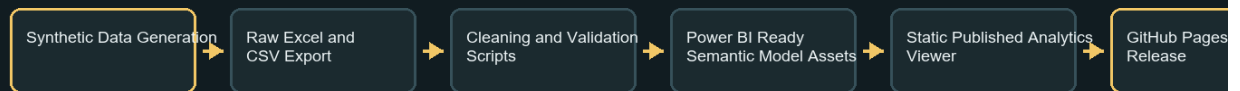


Figure A6. System architecture diagram.

Figure 7: Data Processing Flowchart

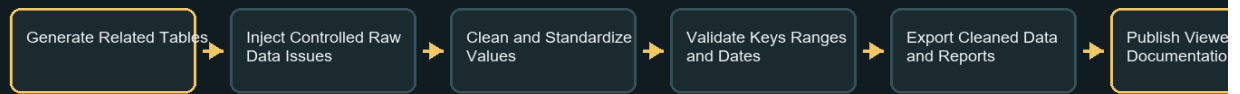
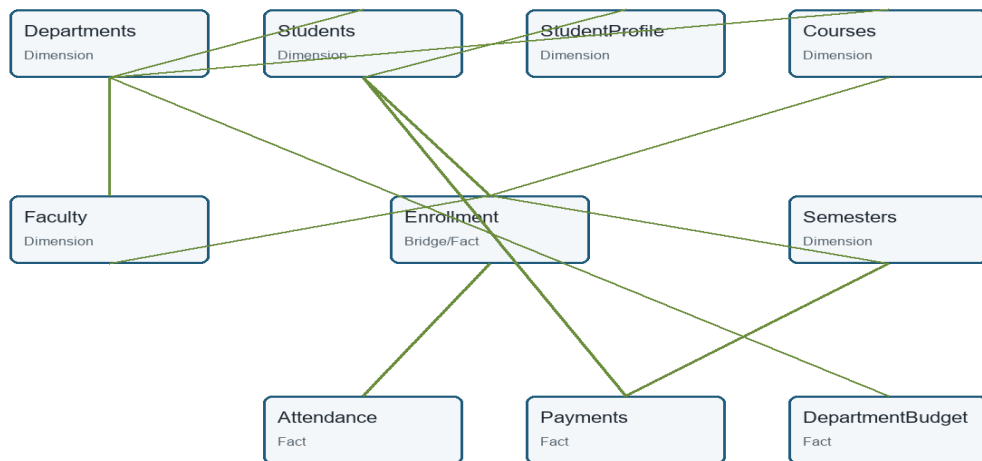


Figure A7. Data-processing workflow.



Conceptual many-to-many: Students -> Enrollment <- Courses. Implement with single-direction relationships where possible.

Figure A8. Entity relationship and data model diagram.